

App Flow and State Machine

Offline Field Issue Tracker

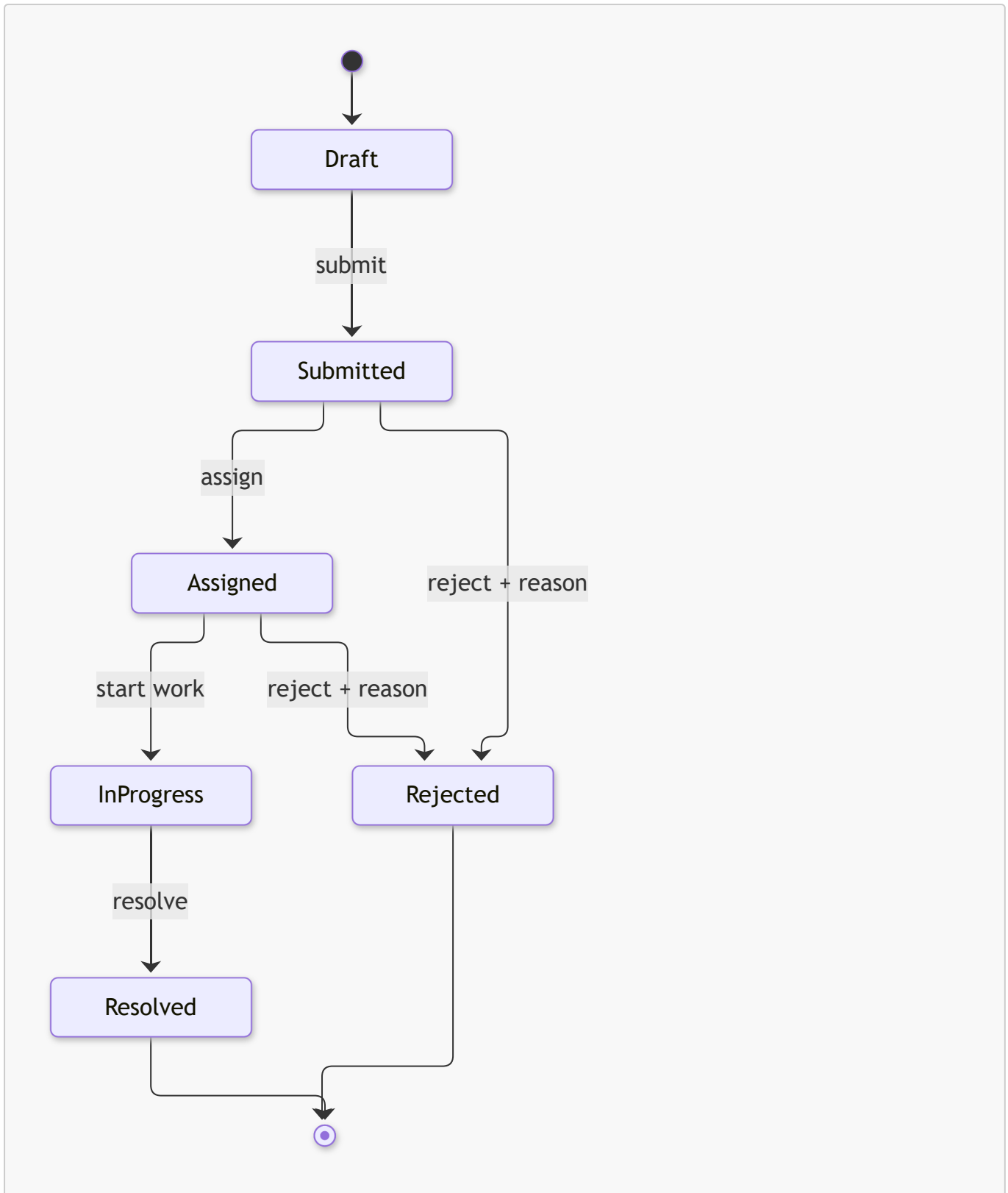
1. Report Lifecycle FSM

States

State	Meaning	Terminal?
Draft	Local composition not submitted	No
Submitted	Server accepted report and it awaits triage	No
Assigned	Coordinator assigned ownership	No
In Progress	Work has started	No
Resolved	Work is reported complete	Yes for the current lifecycle
Rejected	Coordinator rejected the report with a reason	Yes for the current lifecycle

Allowed transitions

From	To	Required actor	Required data
Draft	Submitted	Field Worker	Valid report payload
Submitted	Assigned	Coordinator	Assignee
Assigned	In Progress	Coordinator/assignee	Optional note
In Progress	Resolved	Coordinator/assignee	Resolution note
Submitted	Rejected	Coordinator	Non-empty rejection reason
Assigned	Rejected	Coordinator	Non-empty rejection reason



2. Transition Rules

1. The server is authoritative; the client may display an optimistic state but cannot authorize a transition.
2. The request includes **expected_status**; a mismatch returns 409.
3. The transition endpoint validates actor role, current state, required note, and target state before mutation.
4. Invalid transitions return 409 with **currentStatus**, **requestedStatus**, and **allowedTransitions**.
5. No **report_history** row is written for a rejected transition attempt; an operational sync log may record the attempt.

6. Accepted transitions update `reports.status`, `reports.updated_at`, and insert one history row in one transaction.
7. Status strings are a controlled enum; unknown values are rejected with 422.
8. History is append-only and never rewritten to hide a conflict or correction.

Example invalid response:

```
{
  "error": {
    "code": "INVALID_TRANSITION",
    "message": "Resolved reports cannot transition directly to In Progress",
    "currentStatus": "Resolved",
    "requestedStatus": "In Progress",
    "allowedTransitions": []
  }
}
```

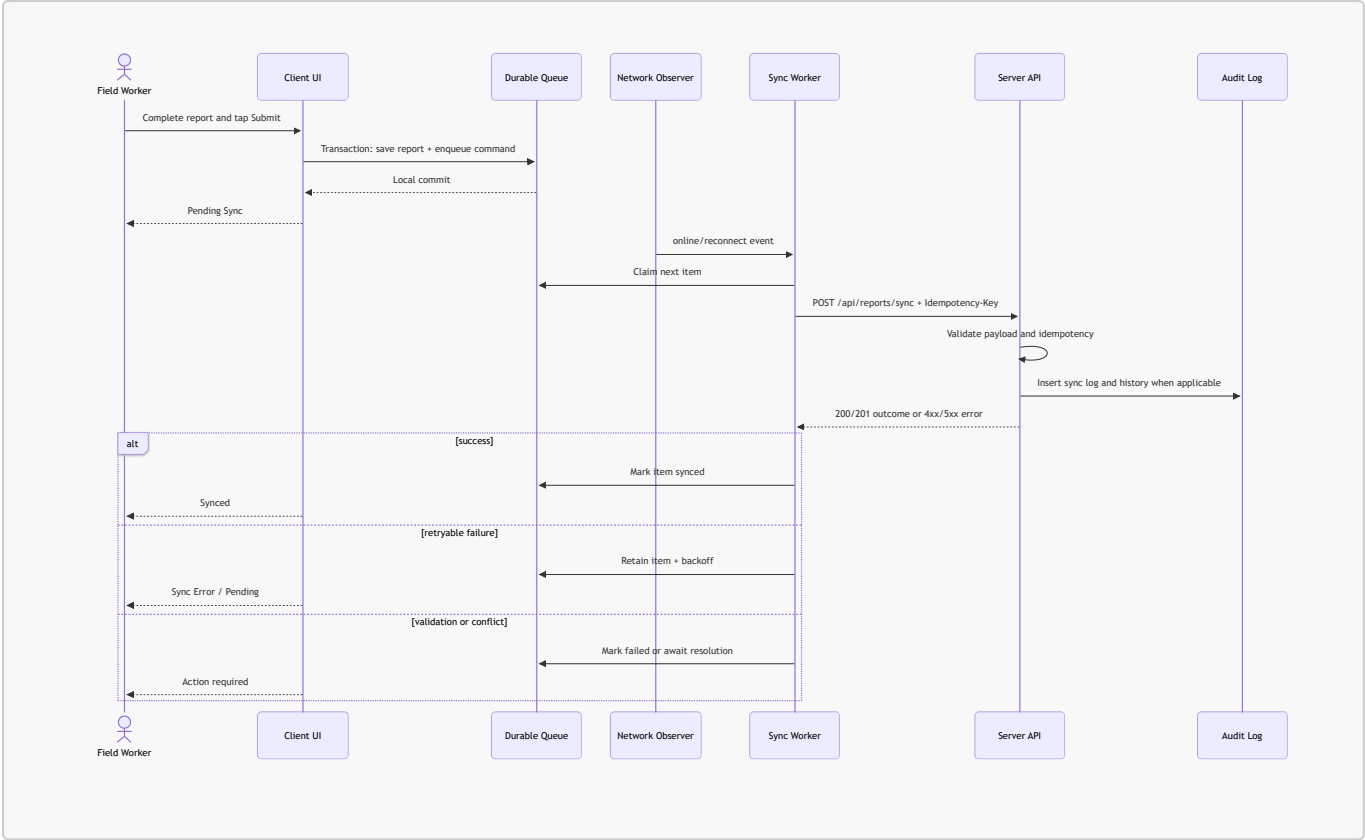
3. Re-opening Resolved Issues

Default behavior is to preserve a resolved report and create a linked follow-up report:

1. Coordinator selects `Create follow-up`.
2. The new UUID is generated and its description references the original report.
3. The new report starts in `Draft` locally or `Submitted` when created by an authorized coordinator.
4. The original remains `Resolved`; its history is unchanged.

An explicit reopen feature may be added only with a product decision. If enabled, `Resolved -> In Progress` must require a reason, coordinator permission, and an audit event recording `reopened_from_resolved=true`. It must never be an implicit side effect of an old client retry.

4. App and Synchronization Flow



5. Offline Queue State

Queue state	Entry condition	Exit
Pending	Local submit committed	Claimed by worker or manual retry
In Flight	Worker owns item	Success, retryable failure, or terminal failure
Retry Scheduled	Network/5xx/429 failure	Backoff timer or reconnect
Failed	422, unrecoverable 409, or permission error	User edits/resolves and retries
Synced	Server acknowledgement	Retention policy cleanup

The worker must use a lease or claim marker to prevent two tabs from processing the same command concurrently. Server idempotency remains required because browser coordination is best-effort.

6. Transition Test Matrix

From	To	Expected
Draft	Submitted	Allow with valid payload
Submitted	Assigned	Allow with coordinator and assignee
Assigned	In Progress	Allow
In Progress	Resolved	Allow with resolution note

From	To	Expected
Submitted	Rejected	Allow with reason
Assigned	Rejected	Allow with reason
Draft	Resolved	Reject 409
Resolved	In Progress	Reject by default; follow-up path
Rejected	Assigned	Reject 409
Submitted	Rejected	Reject 422 when reason missing